MINDGUARD AI: FULL-SCALE MENTAL HEALTH SENTIMENT PLATFORM

Complete Implementation, System Architecture & Operational Guide based on Vishwakarma University Minor Project Paper

Project Title: Sentimental Analysis for Mental Health Detection	**Institution: Vishwakarma University, Pune (Dept. of Computer Engg.)**
Authors: Arnav Phuke, Sushrut Ingle, Bhupesh Bhatia, Saket Radke, Samruddhi Meher	**Project Guide:** Clinical NLP & Cognitive Systems Advisory Board | **A.Y.:** 2026-2027

1. Executive Summary & Research Realization

Early warning signs of mental health conditions frequently hide in informal, unstructured language rather than clinical records. The research paper presents a transformer-based text classification pipeline fine-tuning bert-base-uncased on the Kaggle Sentiment Analysis for Mental Health corpus (~52,680 records) across 7 states: **Normal, Depression, Suicidal, Anxiety, Stress, Bipolar, and Personality disorder**. While the paper listed cloud deployment, web UI, real-time chatbot screening, and SHAP/LIME explainability as *'future work not yet implemented'*, **MindGuard AI transforms the research paper into a fully working, production-grade platform**.

2. End-to-End System Components Built

Inference Engine	backend/model_engine.py	7-class probability calibration, top-3 predictions, risk triage levels, and token-level attention saliency.
REST API Server	backend/api.py	FastAPI endpoints for /api/analyze, /api/chat, /api/batch, /api/benchmarks, and /api/helplines.
Web Application UI	frontend/index.html	Single-page responsive application with Real-time Screener, MindBot Chatbot, Batch CSV Audit, and Paper Charts.
PyTorch Trainer	backend/train_bert.py	Full PyTorch + Transformers fine-tuning script with AdamW (2e-5), linear decay, and 72/8/20 stratified split.
Dataset Generator	backend/dataset_generator.py	Stratified corpus generator mirroring the 52,680 Kaggle dataset distribution across all 7 classes.
Test & Audit Suite	verify_system.py & evaluator.py	Automated end-to-end verification passing 5/5 test suites and producing confusion matrices.

3. Research Paper Empirical Benchmarks (Held-out Test Set N=10,536)

Class	Test Samples	Class Share (%)	Precision	Recall	F1-Score
Normal	3,269	31.0%	0.9576	0.9523	0.9549 (Strongest)
Anxiety	768	7.3%	0.8753	0.8958	0.8855
Bipolar	555	5.3%	0.8330	0.8721	0.8521
Depression	3,081	29.2%	0.7712	0.7790	0.7751
Personality disorder	215	2.0%	0.8093	0.7302	0.7677
Stress	518	4.9%	0.7866	0.7471	0.7663
Suicidal	2,130	20.2%	0.7151	0.7117	0.7134 (Most Challenging)
Overall Test Accuracy	**10,536**	**100.0%**	**Macro F1: 0.8164**	**Weighted F1: 0.8299**	**83.00% (Target: >80%)**

4. Key Innovations & Operational Capabilities

1. Live Real-Time Screener & Saliency Heatmap: Computes top-3 class probabilities and highlights exact tokens that influenced the prediction (e.g. green for normal/coping words, amber for anxiety/stress, red for acute despair).	**2. Automatic Emergency Crisis SOS Protocol:** If Suicidal ideation or self-harm is detected (confidence ≥ 25%), an interactive crisis banner immediately surfaces direct links to Tele-MANAS (14416) and Vandrevala Foundation (9999 666 555).
3. MindBot Conversational Screener: An empathetic multi-turn screening agent that monitors emotional trajectory and recommends psychological first aid and grounding exercises (4-4-6 breathing, DBT grounding).	**4. High-Throughput Batch CSV Screener:** Allows clinical researchers or university counseling centers to upload CSV/TXT files and audit hundreds of student posts simultaneously with doughnut distributions and risk charts.

5. REST API Interface Specification

/api/analyze	POST	{"text": "user statement..."}	top_3_predictions, primary_class, confidence, risk_assessment, token_explainability, crisis_directory
/api/chat	POST	{"messages": [{"role": "user", "content": "..."}]}	reply, message_analysis, distress_flag, triage_recommendation, crisis_directory
/api/batch	POST	Multipart form upload (.csv / .txt)	total_processed, class_distribution, risk_summary, sample_results
/api/benchmarks	GET	None	Paper metadata, Tables I, II, III, IV epoch training dynamics
/api/helplines	GET	None	List of 24x7 verified crisis counseling contacts (India & Global)

6. Execution, Verification & Deployment Instructions

• **1-Click Windows Launch:** Double-click run_mindguard.bat in C:\Users\Nucom\Projects\MindGuard_AI\. The script automatically verifies packages, starts Uvicorn on http://127.0.0.1:8000, and launches your web browser.
• **Automated Verification Suite:** Execute python verify_system.py to run all 5 test suites (7-class inference, crisis triggers, probability calibration, token saliency, and API endpoints).
• **Model Evaluation:** Run python backend/evaluator.py to re-evaluate the model across the benchmark corpus.

7. Ethical AI Screening & Clinical Safety Guardrails

ETHICAL TRIAGE PROTOCOL:
MindGuard AI is an automated screening and triage research tool developed for early detection and research evaluation. It is **NOT a diagnostic medical device and does NOT replace licensed psychiatrists, clinical psychologists, or medical professionals**. Predictions carry statistical uncertainty, particularly between overlapping categories like Depression and Suicidal ideation. Any statement flagged with critical risk automatically triggers direct connection to official government and NGO crisis hotlines.